MATH 3589: Introduction to Financial Mathematics

Course Outline

Autumn 2026

Instructor: Dr. Kenneth Ng
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Office: Room 440, Mathematics Tower (MW 440)
Class Hours and Location: 12:40 - 1:35 pm, Mon, Wed, Fri; Journalism Bldg 270
Office Hours: Monday and Friday, 10:00 - 11:00 am, or by appointment

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Course Description

This course introduces the mathematical foundations of discrete-time asset and derivative pricing, building on the binomial tree model as the core framework. It develops one-period and multi-period pricing models, the principle of no arbitrage, risk-neutral valuation, replicating portfolios, and hedging strategies. It also covers key mathematical tools in financial mathematics, including conditional expectations, martingales, and change of measures. Students may continue with MATH 5635: Stochastic Calculus for Finance I, for continuous-time pricing theory.

Course Objectives

Upon successful completion of the course, students will be able to:

- Understand fundamental stochastic processes and their key properties, including random walks and martingales.
- Develop and analyze the binomial asset pricing model for financial assets and derivatives.
- Apply no-arbitrage principles and construct replicating portfolios for pricing and hedging.
- Understand and apply risk-neutral pricing and equivalent martingale measures.

- Price and hedge European, American, and path-dependent options, including Asian options.
- Apply binomial models to interest-rate modeling and fixed-income securities.

Course Materials

- The class notes available on Canvas would be the main source of course materials.
- Reference books:
  1. *Stochastic Calculus for Finance I: The Binomial Asset Pricing Model* by Steven Shreve.
  2. *Options, Futures, and Other Derivatives* by John C. Hull.
  3. *Stochastic Calculus for Finance II: Continuous Time Models* by Steven Shreve.
  4. *Probability Essentials* by Jean Jacod and Philip Protter.
  5. *Probability: Theory and Examples* by Rick Durrett.

Prerequisites

Prereq: C- or better in 3345 or credit for 345; and either C- or better in 4530, 5530H, or STAT 4201; or credit for 530, 531H, or STAT 420; or permission of department. Not open to students with credit for 589.

Grading and Assessments

- Grade composition:
  1. Homework: 20% of the final grade. One homework per chapter.
  2. Two midterm exams: 40% of the final grade, each carries 20%.
  3. Final exam: 40% of the final grade.
- Grades of this course will be assigned mainly based on the OSU Standard Grade Scheme. Mild adjustment could be made based on difficulties of the exam papers. For students' reference, the OSU Standard Grade Scheme is given below:

Grade	Score
A	93%
A-	90%
B+	87%
B	83%
B-	80%
C+	77%
C	73%
C-	70%
D+	67%
D	60%
E	Below 60%

Course Policies

Homework Submission Policies

Please submit your assignment as a single PDF file to Canvas. The due dates of homework will be specified on Canvas, which will usually be two weeks after the homework was posted.

Late assignments will be accepted for no penalty if a valid excuse is communicated to the instructor before the deadline. Assignments will be accepted with a 10%/30%/50% deduction within 1/2/3 days after the deadline. After this any assignments handed will be given 0.

Academic Integrity and Honesty

Academic integrity is essential to maintaining an environment that fosters excellence in teaching, research, and other educational and scholarly activities. Thus, The Ohio State University and the Committee on Academic Misconduct (COAM) expect that all students have read and understand the University's Code of Student Conduct, and that all students will complete all academic and scholarly assignments with fairness and honesty. Students must recognize that failure to follow the rules and guidelines established in the University's Code of Student Conduct and this syllabus may constitute Academic Misconduct.

The Ohio State University's Code of Student Conduct (Section 3335-23-04) defines academic misconduct as: Any activity that tends to compromise the academic integrity of the University or subvert the educational process. Examples of academic misconduct include (but are not limited to) plagiarism, collusion (unauthorized collaboration), copying the work of another student, and possession of unauthorized materials during an examination. Ignorance of the University's Code of Student Conduct is never considered an excuse for academic misconduct, so please review the Code of Student Conduct and, specifically, the sections dealing with academic misconduct.

If an instructor suspects that a student has committed academic misconduct in this course, the instructor is obligated by University Rules to report those suspicions to the Committee on Academic Misconduct. If COAM determines that a student violated the University's Code of Student Conduct (i.e., committed academic misconduct), the sanctions for the misconduct could include a failing grade in the course and suspension or dismissal from the University.

If students have questions about the above policy or what constitutes academic misconduct in this course, they should contact the instructor.

Artificial Intelligence and Academic Integrity

There has been a significant increase in the popularity and availability of a variety of generative artificial intelligence (AI) tools, including ChatGPT, Sudowrite, and others. These tools will help shape the future of work, research and technology, but when used in the wrong way, they can stand in conflict with academic integrity at Ohio State.

All students have important obligations under the Code of Student Conduct to complete all academic and scholarly activities with fairness and honesty. Our professional students also have the responsibility to uphold the professional and ethical standards found in their respective academic honor codes. Specifically, students are not to use unauthorized assistance in the laboratory, on field work, in scholarship, or on a course assignment unless such assistance has been authorized specifically by the course instructor. In addition, students are not to submit their work without acknowledging any word-for-word use and/or paraphrasing of writing, ideas or other work that is not your own. These requirements apply to all students undergraduate, graduate, and professional.

To maintain a culture of integrity and respect, these generative AI tools should not be used in the completion of course assignments unless an instructor for a given course specifically authorizes their use. Some instructors may approve of using generative AI tools in the academic setting for specific goals. However, these tools should be used only with the explicit and clear permission of each individual instructor, and then only in the ways allowed by the instructor.

Accommodations for Disabilities

The university strives to maintain a healthy and accessible environment to support student learning in and out of the classroom. If you anticipate or experience academic barriers based on your disability (including mental health, chronic, or temporary medical conditions), please let me know immediately so that we can privately discuss options. To establish reasonable accommodations, I may request that you register with Student Life Disability Services (SLDS). After registration, make arrangements with me as soon as possible to discuss your accommodations so that they may be implemented in a timely fashion.

If you are ill and need to miss class, including if you are staying home and away from others while experiencing symptoms of a viral infection or fever, please let me know immediately. In cases where illness interacts with an underlying medical condition, please consult with Student Life Disability Services to request reasonable accommodations. You can connect with them at slds@osu.edu; 614-292-3307; or https://slds.osu.edu.

Religious Accommodations

Ohio State has had a longstanding practice of making reasonable academic accommodations for students' religious beliefs and practices in accordance with applicable law. In 2023, Ohio State updated its practice to align with new state legislation. Under this new provision, students must be in early communication with their instructors regarding any known accommodation requests for religious beliefs and practices, providing notice of specific dates for which they request alternative accommodations within 14 days after the first instructional day of the course. Instructors in turn shall not question the sincerity of a student's religious or spiritual belief system in reviewing such requests and shall keep requests for accommodations confidential.

With sufficient notice, instructors will provide students with reasonable alternative accommodations with regard to examinations and other academic requirements with respect to students' sincerely held religious beliefs and practices by allowing up to three absences each semester for the student to attend or participate in religious activities. Examples of religious accommodations can include, but are not limited to, rescheduling an exam, altering the time of a student's presentation, allowing make-up assignments to substitute for missed class work, or flexibility in due dates or research responsibilities. If concerns arise about a requested accommodation, instructors are to consult their tenure initiating unit head for assistance.

A student's request for time off shall be provided if the student's sincerely held religious belief or practice severely affects the student's ability to take an exam or meet an academic requirement and the student has notified their instructor, in writing during the first 14 days after the course begins, of the date of each absence. Although students are required to provide notice within the first 14 days after a course begins, instructors are strongly encouraged to work with the student to provide a reasonable accommodation if a request is made outside the notice period. A student may not be penalized for an absence approved under this policy.

If students have questions or disputes related to academic accommodations, they should contact their course instructor, and then their department or college office. For questions or to report discrimination or harassment based on religion, individuals should contact equity@osu.edu

Additional Policies

For more information on university policies, consult https://ugeducation.osu.edu/academics/syllabus-policies-statements/standard-syllabus-statements

Course Syllabus, Schedule, Final Exam

Depending on the teaching progress, the course is tentatively designed to have 6 chapters.

Chapter	Content	References
1 Probability and Stochastic Processes	• Countable probability spaces • Random variables • Stochastic processes • Conditional expectations • Markov processes	[1] Ch 2, [4] Ch 23
2 One-Period Binomial Models	• One-period models • Basics of derivatives • Replicating portfolio and risk-neutral pricing	[1] Ch 1
3 Multi-period Binomial Models	• n-period binomial trees • Replicating portfolio • Risk-neutral pricing formula • Change of measures • Martingales	[1] Ch 2, 3
4 American and Asian Options	• Stopping times • Pricing of American options by backward induction • Path-dependent options	[1] Ch 4
5 Binomial Interest Rate Models Barrier Options	• Binomial model for interest rates • Zero-coupon bonds • Forward measures	[1] Ch 6
6 Random Walk	• Properties of random walks • The central limit theorem • Convergence to continuous time model • Brownian motions and Black-Scholes equation	[1] Ch 4

Schedule

The following table lists the tentative schedule of the course. Students may expect deviations from the table according to actual teaching progress.

Week	Dates	Topics	HW/Exam
1	8/24 - 8/28	Probability and Stochastic Processes	–
2	8/31 - 9/4	Probability and Stochastic Processes	HW1
3	9/7 - 9/11	The One-period Binomial Model	–
4	9/14 - 9/18	The One-period Binomial Model	HW2
5	9/21 - 9/25	The Multi-period Binomial Model	–
6	9/28 - 10/2	The Multi-period Binomial Model / Midterm I	MT1
7	10/5 - 10/9	The Multi-period Binomial Model	–
8	10/12 - 10/16	The Multi-period Binomial Model / Autumn Break	HW3
9	10/19 - 10/23	American and Asian Options	–
10	10/26 - 10/30	American and Asian Options	HW4
11	11/2 - 11/6	Binomial Models for Interest Rate	–
12	11/9 - 11/13	Binomial Models for Interest Rate / Midterm II	MT2
13	11/16 - 11/20	Binomial Models for Interest Rate	HW5
14	11/23 - 11/27	Random Walk / Thanksgiving Break	–
15	11/30 - 12/4	Random Walk	HW6
16	12/7 - 12/9	Final Review	–

Exam Date

- Midterm 1: 10/2 (Fri), during class time
- Midterm 2: 11/13 (Fri), during class time
- Final: 12/17 (Thur), 2:00pm-3:45pm, Journalism Bldg 270 (usual classromm)